

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40013
Subject Title	Energy Conversion and Storage with Machine Learning
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: Material Science (AP20002) Exclusion: Materials In Energy Conversion And Storage (AP40011)
Objectives	To introduce the concepts, principles and materials applied in the latest technologies for energy conversion and storage processes. To introduce basic concepts and techniques used in machine learning for the design of materials and management of device systems for energy conversion and storage.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) understand the basic working principles and processes of energy conversion and storage in capacitors, batteries, solar cells, and heterogeneous catalysis; b) identify key materials used in capacitors, batteries, solar cells, and heterogeneous catalysis, and solve corresponding problems associated with those materials; and c) identify research problems that can be solved by machine learning based on judiciously selected datasets from materials or devices for energy conversion and storage.
Subject Synopsis/ Indicative Syllabus	Materials for Energy Conversion and Storage: Physical principles of and key materials for solar cells, batteries, capacitors and heterogeneous catalysis. Machine Learning in Energy Storage Materials and Devices: Machine learning examples in materials design for solar cells, batteries, capacitors and heterogeneous catalysis; Machine learning examples in lifetime prediction and management of devices.
Teaching/Learning Methodology	Lecture: The working principles of various kinds of energy conversion and storage processes will be explained, with particular emphasis on the key material issues for the latest development in solar cells, batteries, supercapacitors and heterogeneous catalysis. Examples will be used to illustrate how the basic concepts of machine learning can be applied to the design of materials and the management of device systems for better performance in energy conversion and storage processes. Tutorial: Students will work on a prescribed set of problems in tutorials. They are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets

	provide them opportunities to apply their knowledge gained from the lecture. They also help the students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in relation to daily life phenomena or experience.																										
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th></tr> <tr> <td>1. Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>2. Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100</td><td></td><td></td><td></td></tr> </table>				Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			a	b	c	1. Continuous assessment	40	✓	✓	✓	2. Examination	60	✓	✓	✓	Total	100			
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	Total	100																									
Continuous assessment: The continuous assessment includes assignments, oral presentation, and mid-term test which aim at checking the progress of student study throughout the course, assisting them in fulfilling the learning outcomes. Assignments, in general, are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. One oral presentation is used to train the students with the ability to extract useful information from various sources and the ability to present in a clear, logic, and well-organized way. One mid-term test will be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended learning outcomes, and as a means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It is a closed-book examination. The emphasis of assessment is to test the understanding, analysis and problem solving ability of the students.																											
Student Study Effort Expected	Class contact:																										
	▪ Lecture & Tutorial		39 Hrs.																								
	Other student study effort:																										
	▪ Self-study		81 Hrs.																								
	Total student study effort		120 Hrs.																								
Reading List and References	Kathy Lu, Materials in Energy Conversion, Harvesting and Storage, John Wiley & Sons, 2014. C.D.Rahn and C.-Y.Wang, Battery Systems Engineering, John Wiley & Sons, 2013.																										

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